

Local and Global Quantities for One-Dimensional Linear Elements

ME/BME 736 – Advanced Finite Element Methods

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This handout uses one three-element mesh to explain the distinction between local and global quantities and how element load vectors and stiffness matrices are assembled. The notation is the same as in Chapter 5.

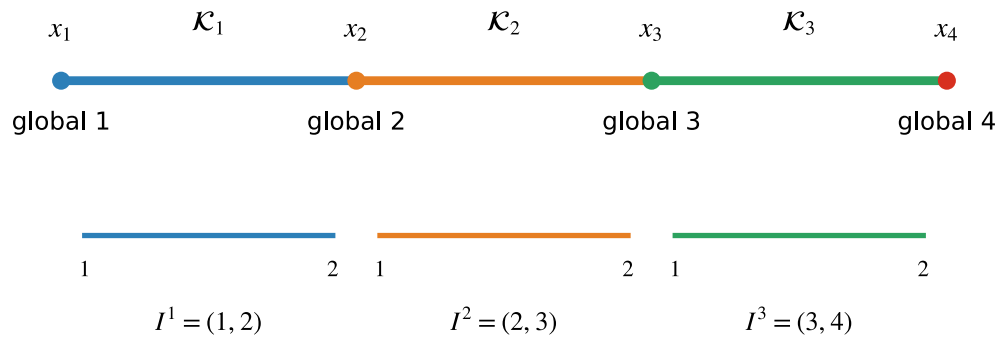
1 One mesh with local and global node numbers

Let

$$0 = x_1 < x_2 < x_3 < x_4 = L$$

be the global nodes of a mesh containing three elements:

$$\mathcal{K}_1 = (x_1, x_2), \quad \mathcal{K}_2 = (x_2, x_3), \quad \mathcal{K}_3 = (x_3, x_4).$$



The global node numbers identify nodes in the complete mesh. The local node numbers restart at 1 and 2 on every element. Thus the same physical node x_2 is local node 2 of element 1 and local node 1 of element 2.

1.1 The three maps used in assembly

The local-to-global map I^e tells us the global number of a local node:

e	$I^e(1)$	$I^e(2)$
1	1	2
2	2	3
3	3	4

 $\implies I^e(1) = e, \quad I^e(2) = e + 1.$

The inverse map J^e is used only on element e . If global node i belongs to element e , then $J^e(i)$ gives its local number on that element. For example,

$$J^1(2) = 2, \quad J^2(2) = 1.$$

The superscript is necessary because global node 2 has a different local number in the two elements that meet there.

Finally, let $\mathcal{E}(i)$ denote the elements attached to global node i . For this mesh,

$$\mathcal{E}(1) = \{1\}, \quad \mathcal{E}(2) = \{1, 2\}, \quad \mathcal{E}(3) = \{2, 3\}, \quad \mathcal{E}(4) = \{3\}. \quad (1)$$

Thus

$$\begin{aligned} I^e(a) &: \text{local node } a \text{ goes to which global node?} \\ J^e(i) &: \text{global node } i \text{ has which local number on element } e? \\ \mathcal{E}(i) &: \text{which elements meet at global node } i? \end{aligned}$$

2 A global finite element function on one element

Let $\{\phi_i\}_{i=1}^4$ be the global nodal basis. The basis function ϕ_i has value one at global node i and zero at every other global node:

$$\phi_i(x_j) = \delta_{ij}. \quad (2)$$

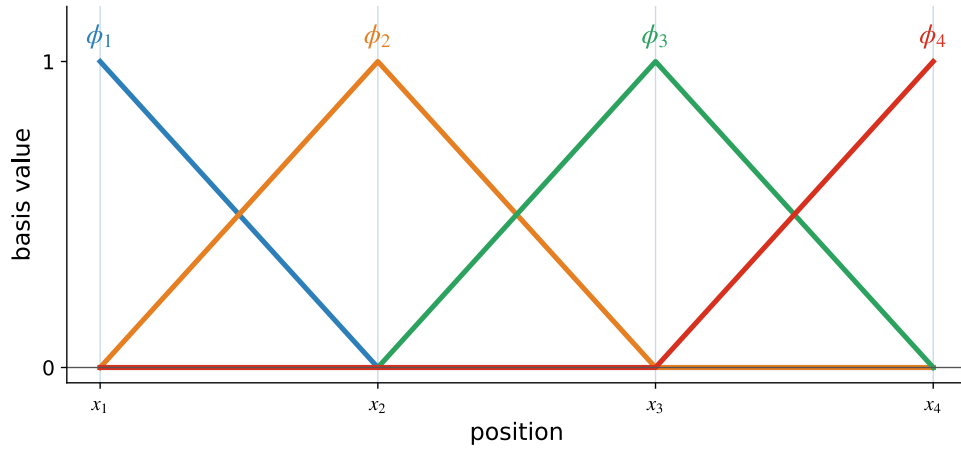
The support of ϕ_i is defined by

$$\text{supp } \phi_i = \overline{\{x \in [0, L] : \phi_i(x) \neq 0\}}. \quad (3)$$

For the linear nodal basis, this support is the union of the closed elements attached to node i :

$$\text{supp } \phi_i = \bigcup_{e \in \mathcal{E}(i)} \bar{\mathcal{K}}_e. \quad (4)$$

Thus an interior basis function is supported on the two elements that meet at its node, while an endpoint basis function is supported on one element.



A finite element function is written globally as

$$u_h(x) = \sum_{i=1}^4 U_i \phi_i(x). \quad (5)$$

The global degree of freedom U_i is the nodal value $u_h(x_i)$.

On element $\mathcal{K}_e$, only the basis functions associated with its two nodes are nonzero. Number the two local nodes from left to right and define their coordinates by

$$x_1^e = x_e, \quad x_2^e = x_{e+1}. \quad (6)$$

The corresponding local degrees of freedom are

$$U_1^e = u_h(x_1^e), \quad U_2^e = u_h(x_2^e). \quad (7)$$

Let ϕ_1^e and ϕ_2^e be the local basis functions:

$$\phi_1^e(x) = \frac{x_{e+1} - x}{h_e}, \quad \phi_2^e(x) = \frac{x - x_e}{h_e}, \quad h_e = x_{e+1} - x_e. \quad (8)$$

They satisfy $\phi_a^e(x_b^e) = \delta_{ab}$. Consequently, the restriction of u_h to element e is first written entirely in local quantities as

$$u_h|_{\mathcal{K}_e} = U_1^e \phi_1^e + U_2^e \phi_2^e. \quad (9)$$

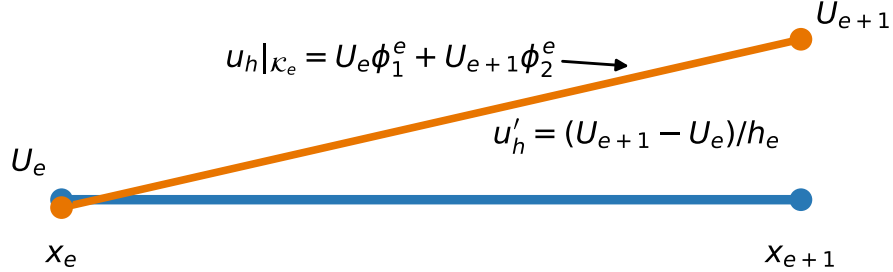
The local-to-global map identifies the same nodes in global notation:

$$U_a^e = U_{I^e(a)}, \quad \phi_a^e = \phi_{I^e(a)}|_{\mathcal{K}_e}, \quad a = 1, 2. \quad (10)$$

Substituting these relations into Equation (9) gives

$$u_h|_{\mathcal{K}_e} = \sum_{a=1}^2 U_{I^e(a)} \phi_a^e = U_{I^e(1)} \phi_1^e + U_{I^e(2)} \phi_2^e. \quad (11)$$

For the consecutive numbering used here, $I^e(1) = e$ and $I^e(2) = e + 1$, so the two coefficients in Equation (11) are U_e and U_{e+1} .



For example, $I^2 = (2, 3)$, and therefore

$$u_h|_{\mathcal{K}_2} = U_2 \phi_1^2 + U_3 \phi_2^2, \quad \mathbf{U}^2 = \begin{bmatrix} U_2 \\ U_3 \end{bmatrix}. \quad (12)$$

For all three elements, the local coefficient vectors are

$$\mathbf{U}^1 = \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}, \quad \mathbf{U}^2 = \begin{bmatrix} U_2 \\ U_3 \end{bmatrix}, \quad \mathbf{U}^3 = \begin{bmatrix} U_3 \\ U_4 \end{bmatrix}.$$

3 Assembly of the load vector

For a distributed load f , the global load entry is

$$F_i = \int_0^L f \phi_i \, dx. \quad (13)$$

Splitting the integral over all three elements first gives

$$F_i = \sum_{e=1}^3 \int_{\mathcal{K}_e} f \phi_i \, dx. \quad (14)$$

If element e does not belong to $\mathcal{E}(i)$, then $\phi_i|_{\mathcal{K}_e} = 0$, and its integral in Equation (14) is zero. The full sum therefore reduces to

$$F_i = \sum_{e \in \mathcal{E}(i)} \int_{\mathcal{K}_e} f \phi_i \, dx. \quad (15)$$

Define the two local load entries on element e by

$$F_a^e = \int_{\mathcal{K}_e} f \phi_a^e \, dx, \quad a = 1, 2. \quad (16)$$

Because $\phi_i|_{\mathcal{K}_e} = \phi_{J^e(i)}^e$, Equation (15) becomes

$$\boxed{F_i = \sum_{e \in \mathcal{E}(i)} F_{J^e(i)}^e.} \quad (17)$$

For the three-element mesh,

$$F_1 = F_1^1, \quad F_2 = F_2^1 + F_1^2, \quad F_3 = F_2^2 + F_1^3, \quad F_4 = F_2^3.$$

Thus the global load vector is the sum of the three element vectors placed in their global positions:

$$\boxed{\mathbf{F} = \begin{bmatrix} F_1^1 \\ F_2^1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ F_1^2 \\ F_2^2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ F_1^3 \\ F_2^3 \end{bmatrix} = \begin{bmatrix} F_1^1 \\ F_2^1 + F_1^2 \\ F_2^2 + F_1^3 \\ F_2^3 \end{bmatrix}.} \quad (18)$$

At a shared node, the contributions from the neighboring elements are added.

4 Assembly of the stiffness matrix

The global stiffness entry for the axial bar is

$$K_{ij} = \int_0^L AE \phi_j' \phi_i' \, dx. \quad (19)$$

Define the local stiffness entries on element e by

$$K_{ab}^e = \int_{\mathcal{K}_e} AE (\phi_b^e)' (\phi_a^e)' \, dx, \quad a, b = 1, 2. \quad (20)$$

An element contributes to K_{ij} only when it contains both global nodes i and j . Hence

$$\boxed{K_{ij} = \sum_{e \in \mathcal{E}(i) \cap \mathcal{E}(j)} K_{J^e(i), J^e(j)}^e.} \quad (21)$$

For example,

$$K_{22} = K_{22}^1 + K_{11}^2, \quad K_{23} = K_{12}^2, \quad K_{14} = 0.$$

Let

$$\mathbf{K}^e = \begin{bmatrix} K_{11}^e & K_{12}^e \\ K_{21}^e & K_{22}^e \end{bmatrix}.$$

Placing each 2×2 element matrix into the rows and columns specified by I^e gives

$$\begin{aligned} \mathbf{K} &= \begin{bmatrix} K_{11}^1 & K_{12}^1 & 0 & 0 \\ K_{21}^1 & K_{22}^1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & K_{11}^2 & K_{12}^2 & 0 \\ 0 & K_{21}^2 & K_{22}^2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & K_{11}^3 & K_{12}^3 \\ 0 & 0 & K_{21}^3 & K_{22}^3 \end{bmatrix} \\ &= \begin{bmatrix} K_{11}^1 & K_{12}^1 & 0 & 0 \\ K_{21}^1 & K_{22}^1 + K_{11}^2 & K_{12}^2 & 0 \\ 0 & K_{21}^2 & K_{22}^2 + K_{11}^3 & K_{12}^3 \\ 0 & 0 & K_{21}^3 & K_{22}^3 \end{bmatrix}. \end{aligned} \quad (22)$$

For a linear bar element with constant AE ,

$$\mathbf{K}^e = \frac{(AE)_e}{h_e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \quad (23)$$

If $k_e = (AE)_e/h_e$, the assembled matrix is

$$\mathbf{K} = \begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix}. \quad (24)$$

5 The assembly rule

For every element e , compute its two local load entries and its four local stiffness entries. Then use I^e to add them to the global vector and matrix:

$$\boxed{F_{I^e(a)} += F_a^e, \quad K_{I^e(a), I^e(b)} += K_{ab}^e.} \quad (25)$$

The corresponding algorithm is

```
initialize F = 0 and K = 0
for each element e = 1, ..., N:
    compute F^e and K^e
    for a = 1, 2:
        i = I^e(a)
        F_i += F_a^e
        for b = 1, 2:
            j = I^e(b)
            K_ij += K_ab^e
```

In Python, subtract one from every mathematical node and element number when using an array index. The mathematical derivation itself remains one-based.